

Instruct Models and Post-Training

Instruct Language Models

- The original definition of a language model was a probabilistic model that defines a distribution over finite strings, given a vocabulary
 - Equivalently, the prediction of the next word (token) given a prefix
- Many tasks can be cast as (iterative) next word prediction
 - Therefore, effectively performing next word prediction may allow zero-shot or few-shot learning for many tasks
 - OpenAI, with their GPT3 model (Brown et al., 2020), has shown that optimizing for next word prediction yields a strong model for few-shot and zero-shot learning

Instruct Language Models

- Instruct language models have redefined the notion of a language model to a system that obeys instructions
 - Initially, instructions to produce text
 - More recently, instructions to perform actions (agents)
- Instruct models are based on pretrained, next word prediction models that serve as initialization to a subsequent stage, sometimes called post-training

Instruction Following vs. Next Word Prediction

Main differences:

1. Instruction following is an interaction → requires adherence to the implied intent of the user
2. Instruction following usually imposes safety constraints / guardrails
3. Next word prediction is a density estimation problem → we seek to give probability to all "good" completions

Instruction Following vs. Next Word Prediction

Most medieval European scholars believed that the Earth was ...

meta-llama/Llama-3.1-8B

- flat, and they believed it was flat because they
- a flat disk, surrounded by water
- flat
- at the center of the universe, but this belief
- at the centre of the universe and that the stars

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Most medieval European scholars believed that the Earth was ****at the center of the universe****, with the Sun, Moon, planets, and stars revolving around it. This view is known as the ****geocentric model**** and was based largely on the ideas of the ancient Greek philosopher Ptolemy, whose work ***Almagest*** was highly influential throughout the Middle Ages.